

INTERSHIP REPORT

AWS in an Online Examination System

Introduction

Today's internship session was about **Amazon Web Services (AWS)**. Instead of learning only the theoretical concepts of AWS services, I explored how they can be applied in a real-world project. I selected an **Online Examination System** because educational institutions need to conduct secure online exams, manage student registrations, store question papers, evaluate results, and provide instant feedback. AWS offers cloud services that make the system secure, scalable, reliable, and easy to manage.

Why Cloud Computing?

If an institution manages its online examination system using its own servers, it must handle hardware maintenance, software updates, backups, storage, and security. These tasks require significant time and cost. By using AWS cloud services, the institution can focus on conducting examinations while AWS manages the infrastructure. During examination periods, the system can automatically scale to support thousands of students and pay only for the resources it uses.

Amazon EC2

Amazon EC2 can be used to host the Online Examination System. Students, teachers, and administrators can securely access the platform through EC2 servers. During peak hours, such as competitive exams or university examinations, additional EC2 instances can be added to handle increased traffic. This ensures smooth exam sessions without delays or server failures.

AWS Lambda

AWS Lambda is useful for performing automatic background tasks. For example, Lambda can automatically send exam registration confirmations, exam reminders, OTP verification, and result notifications. It can also calculate objective-type exam scores immediately after submission and generate digital scorecards. Since Lambda runs only when required, it helps reduce operational costs.

Amazon S3

The Online Examination System generates digital files such as question papers, answer sheets, admit cards, result reports, and certificates. These files can be securely stored in Amazon S3. S3 provides highly durable storage, automatic backups, and easy access for authorized users. Older records can also be moved to lower-cost storage classes for long-term storage.

Database Services

Student information, exam schedules, question banks, answer records, results, and attendance details can be stored in **Amazon RDS**. If the system requires very fast access to frequently used data, such as live exam sessions or instant result processing, **Amazon DynamoDB** can also be used. Managed database services improve reliability, availability, and reduce maintenance work.

IAM and Security

AWS Identity and Access Management (IAM) provides secure access control for different users. Students can access only their own exams and results, teachers can create question papers and evaluate answers, and administrators can manage the complete examination system. IAM helps protect confidential exam data through role-based access control and strong security policies.

Load Balancer and Reliability

Large online examinations may have thousands of students accessing the system at the same time. An **AWS Load Balancer** distributes incoming requests across multiple EC2 instances to maintain smooth performance and prevent server overload. AWS monitoring and backup services help reduce downtime and ensure that exam data remains safe and available even during unexpected failures.

Conclusion

From today's internship session, I understood that AWS is not only useful for technology companies but also for educational applications such as Online Examination Systems. Services such as **EC2, Lambda, S3, RDS, IAM, and Load Balancer** work together to build a secure, scalable, and reliable examination platform. This practical example helped me understand how AWS services solve real-world challenges in conducting online examinations instead of simply learning their definitions.